Open problems:
Projective measurements are equilibration processes

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Unbounded observables.** For unbounded X the Lindblad equation of EQ-LIND holds only on a domain; the brief claims that the blockwise solution (5) "extends the resulting semigroup to every density operator" and that $\rho(t)$ stays a density operator. Neither the extension nor Theorem 1 for unbounded X is proved. (Brief, Sec. 3, after eq. (5).)

2. **Non-commutative Scheffé lemma.** "Hilbert–Schmidt convergence to a density operator implies trace-norm convergence" for density operators is cited without reference or proof. The formalization bypasses it, so the lemma itself remains unproved here. (Brief, Sec. 3, between eqs. (9) and (10).)

3. **Continuous spectrum.** The claim that all four conditions fail on a continuous part of the spectrum of X: an operator commuting with X and supported on a non-atomic spectral sector "cannot be a non-zero trace-class operator", and Lindblad fixed points there are not density operators. Not proved. (Brief, Sec. 4, "Observables with continuous spectra".)

4. **A formalism with finite entropy.** When $S(P_X(\rho)) = +\infty$, EQ-KL does not determine a closest ensemble. The brief suggests that "the real problem is the Hilbert space formalism, as it does not guarantee finite entropy for all density operators"; a state space in which every ensemble has finite entropy is not given. (Brief, Sec. 4, "Relative entropy in infinite dimensions".)

5. **Other dynamics with the same limit.** "Other dynamics may have the same asymptotic map": which Lindblad (or other) dynamics have $\mathcal{P}_X$ as their long-time limit, and whether physical measurements are equilibrations "that happen at a fast scale", is left open. (Brief, Sec. 4, "Equilibration as a black-box description"; Sec. 5.)

6. **Repeatability and state update.** The brief claims that repeatability is "intrinsic in requiring that all outputs are equilibria of the process", and that state update on a known outcome "is equivalent to what happens in the classical case". Neither is made precise or proved. (Brief, Sec. 5.)

7. **Destructive measurements.** "The black-box idea could be extended to destructive measurements", with the system state as input and "the final state of the measurement device" as output. No such characterization is given. (Brief, Sec. 5.)

From the formalization

Beyond finite dimension the formalization has a part in any dimension, which covers bounded observables with pure point spectrum only; it leaves open the items "Unbounded observables" and "Continuous spectrum" above and "Bridge from the spectral theorem" below.

1. **Bridge from the spectral theorem.** Observables are given by an eigendecomposition (P_x, x). That a self-adjoint X with pure point spectrum has one, with $P_x = \mu_S(\{x\})$ its spectral projections, is not formalized. (Log, "Hilbert spaces of any dimension", "Differences from the brief".)

2. **Basis independence of the relative entropy.** Lindblad's formula for $D(\rho\|\sigma)$ is shown to be independent of the chosen eigenbases only when $S(\rho) < \infty$; for $S(\rho) = \infty$ this is open. (Log, "Hilbert spaces of any dimension", "Differences from the brief".)

3. **Agreement of the two relative entropies.** That the relative entropy of the any-dimension part agrees with the finite-dimensional `relEntropy` on finite-dimensional spaces is not formalized. (Log, "Hilbert spaces of any dimension", "Differences from the brief".)